C 40625 (Pages : 2) Name.....

Reg. No.....

SIXTH SEMESTER U.G. DEGREE EXAMINATION MARCH 2023

(CBCSS—UG)

Physics/Applied Physics

PHY 6B 10/APH 6B 10—THERMODYNAMICS

(2019 Admission onwards)

Time : Two Hours Maximum : 60 Marks

The symbols used in question paper have their usual meanings.

Section A (Short Answer Type)

Answer **all** *questions in two* **or** *three sentences, each correct answer carries a maximum of* 2 *marks.*

1. Prove that all Carnot engines operating between the same two reservoirs have the same efficiency.
2. What are the features of thermodynamic temperature scale ?
3. Plot the TS diagram of a Carnot's cycle.
4. What are the insights obtained from the relation $dU = đW + đQ$?
5. What is Helmholtz function ? Why is it important ?
6. Derive Clausius theorem.
7. What are the general characteristics of macroscopic co-ordinates ?
8. Distinguish between the systems separated by adiabatic walls and diathermic walls.
9. How is external work different from internal work ?
10. Differentiate between isobaric and isochoric processes.
11. State and explain the second law of thermodynamics.
12. Comment on the molar heat capacities of monatomic gases.

. (Ceiling 20)

Section B (Paragraph/Problem Type)

Answer **all** *questions in a paragraph of about* **half a page** *to* **one page**, *each correct answer carries a maximum of* 5 *marks.*

13. Under what pressure ice freezes at 271 K if the change in specific volume when 1 kg. of water freezes is 91×10^{-6} m^3. Given latent heat of ice = 3.36×10^5 Jkg^{-1}.

14. Prove the principle of increase of entropy.

15. What is a hydrostatic system ? Briefly explain.

16. Show that adiabatics are steeper than isothermals.

17. A mass of mercury at standard atmospheric pressure and a temperature of 25 °C is kept at constant volume. If the temperature is raised to 27°C, what will be the final pressure ? For mercury, volume expansivity = 1.81×10^{-4} K^{-1} and the isothermal compressibility = 4.01×10^{-11} Pa^{-1}.

18. Determine the work done in an adiabatic process in terms of temperature.

19. Derive the relation connecting Cp and Cv.

(Ceiling 30)

Section C (Essay Type)

Essay-Answer in about **two pages**, *any* **one** *questions.*
The question carries 10 *marks.*

20. Discuss the equality of the ideal gas and thermodynamic temperatures.

21. Explain the PV diagram and PT diagram of H_2O.

(1 × 10 = 10 marks)